

A combinatorial CRISPR/enCas12a KO screen to test genetic interactions between Solute Carriers

DOI: 10.5281/zenodo.10354692

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Protocol description

The Solute Carrier (SLC) family is characterized by a high level of substrate redundancy which may diminish phenotypes in regular KO models since lost SLC functions can be compensated by other family members. A systematic screen for genetic interactions, i.e., when the combinatorial KO of two genes shows a stronger phenotype than the added individual KOs, is therefore crucial to reveal otherwise hidden loss-of-function phenotypes.

This protocol describes our effort to use a recently developed combinatorial CRISPR enCas12a KO system to test all possible pairwise interactions for all 258 SLCs that are expressed (>1 transcript per million) in the human colon cancer cell line HCT116. The library for this project contained around 640.000 gRNA combinations and was cloned as four sub-libraries (around 160.000 oligos each) that were used for four consecutive sub-screens. Cell numbers and reactions can be scaled up or down if needed to adapt the protocol for comparable genetic interaction screens.

Materials

Biological materials:	General materials: ▪ 293T packaging cells ▪ pMD2.G (Addgene #12259), psPax2 (Addgene #12260) packaging constructs ▪ enCas12a expressing HCT116 clone (HCT116-enCas12a-c8, RESOLUTE #CE06WS-Q) Optional materials (for library cloning): ▪ Cas12a gRNA expression vector pRDA_052 (Addgene #136474) or modified pRDA_052 vector with two PacI sites (RESOLUTE #PD04N8-9) ▪ Oligo library (recommended: Twist bioscience) ▪ Endura Electrocompetent cells and recovery medium Optional (for generation of enCas12a expressing cell line): ▪ pRDA_174 vector (Addgene #136476)
Reagents:	General reagents: ▪ Q5 Polymerase (NEB) or self-purified Pfu Polymerase (with Phusion HF buffer, NEB #B0518S) ▪ AmpliClean™ Cleanup Kit (NimaGen, SKU: AP-050) Optional (for library cloning): ▪ Esp3I (NEB) ▪ T7 DNA Ligase (NEB) ▪ BSA (10 mg/ml) (NEB) ▪ Dithiothreitol (DTT)
Equipment:	▪ 10-stack tissue culture dishes (Corning, #3270) ▪ Electroporation equipment (optional for custom library) ▪ CASY cell counter

Procedure

Step 1: Library cloning (skip step if using ready-made library)

1.1 Cas12a gRNA design

While usually 4-6 gRNAs per gene should be selected for CRISPR KO screens, we only selected three gRNAs per SLC for this screen to keep the library at a reasonable size. For smaller libraries, we recommend at least 6 gRNAs per gene.

As controls, we chose 12 gRNAs targeting 12 different olfactory receptors (ORs). The three top-scoring gRNAs were selected using the CRISPick tool for enCas12a. (<https://portals.broadinstitute.org/gppx/crispick/public>) (1)(2).

1.2 Oligo design for pooled synthesis

Depending on the desired library, oligos can be synthesized in two ways:

Option 1: order oligos containing both gRNAs:

This option allows flexibility in which gRNAs should be combined since each oligo is designed and synthesized individually:

5'- FW primer – CGTCTCTAGAT – gRNA1 – TAATTTCTACTGTCGTAGAT – gRNA2 – TTTTTGAATCGAGACG – RV primer (reverse-complement) -3'

Option 2: order FW and RV oligos with one gRNA each

In this option all gRNAs on the FW oligos will be recombined with all gRNAs on the RV oligos through the common direct repeat (DR) sequence (TAATTTCTACTGTCGTAGAT) that is present in both FW and RV oligos. However, using different primer sequences for different subsets of oligos allows specific PCR amplification of a subset of gRNA combinations. E.g., we divided the 258 target SLCs into four subgroups and chose four distinct FW oligo primer for each of them. FW oligos with control gRNAs targeting 12 olfactory receptors were designed with all four FW primer sequences. The RV oligos (SLCs and olfactory receptors) all contain the same primer sequence. This strategy allows the amplification of four sub-libraries in four separate PCR reactions, each containing a subset of SLC gRNAs on the 1. gRNA position combined with all other SLCs on the 2. gRNA position.

FW oligo: 5'-FW primer – CGTCTCTAGAT – gRNA1 – TAATTTCTACTGTCGTAGAT -3'

RV oligo: 5'-RV primer – CGTCTCGATTCAAAAAA – gRNA2 (reverse-complement) - ATCTACGACAGTAGAAATTA -3'

Alternatively, if unique gRNAs in each gRNA pair are preferred (gene1-gRNA_1-gene2_gRNA_1, gene1-gRNA_2-gene2_gRNA_2, gene1-gRNA_3-gene2_gRNA_3, etc.), oligos can be designed so each gRNA group (e.g., 1-6) is synthesized with sequences for distinct primer pairs (FW_primer1_gRNA_1 + RV_primer1_gRNA_1, FW_primer2_gRNA_2, RV_primer2_gRNA_2, etc.). That way, each amplified sub-library will only contain one combined gRNA pair from one gRNA group. After amplification, sub-libraries can be mixed at equimolar ratios or clones separately and mixed before virus production.

Examples for FW and RV primer for oligo amplification:

Primer ID	used as FW/RV	primer sequence
GW69	FW	GAGACCATTGCGAATACCAG
GW70	FW	GAACAGTGTGGATGTGACGA
GW71	FW	AGGCCGTTCGTTAATACTTG
GW72	FW	CGGACGGACAGATTACTAGA
GW73	RV	AGAAGGCCGTAGTCAGTTAC

1.3 Amplify ssDNA oligos by PCR:

Option 1: (library oligos containing both gRNAs ordered)

1xMM	
Q5 buffer	10 µl
NTPs (10 mM)	1 µl
Q5 Pol	0.5 µl
oligo amplification primer (10 µM)	2.5 µl
Cas12a oligos (0.5 ng/µl)	5 ng
H2O	to 47.5 µl

- Set up PCR amplification reaction:

Step1: 98°C for 30 sec

Step2: use recommended annealing temperature, e.g., 64°C (GW69+GW73), 65°C (GW70+GW73), 63°C (GW71+GW73) and 64°C (GW72+GW73) for 30 sec

Step3: 72°C for 15 sec

Repeat Step2-3 for 12-16 cycles

Keep at 4°C

Option 2: (library ordered as FW and RV oligos with one gRNA each)

- Set up strand extension reaction:

1xMM	
Q5 buffer	10 µl
NTPs (10 mM)	1 µl
Q5 Pol	0.5 µl
Cas12a oligos FW/RV (0.5 ng/µl)	5 ng
H2O	to 47.5 µl

- Run program in thermocycler:

Step1: 98°C for 30 sec

Step2: 98°C for 10 sec

Step3: 55°C for 30 sec

Step4: 72°C for 10 sec

Repeat Step2-4 for 5 cycles

Cool down to 4°C

- Add 2.5 µl oligo amplification primer pairs (10 µM) to strand extension reaction

- Set up PCR amplification reaction:

Step1: 98°C for 30 sec

Step2: use recommended annealing temperature, e.g., 64°C (GW69+GW73), 65°C (GW70+GW73), 63°C (GW71+GW73) and 64°C (GW72+GW73) for 30 sec

Step3: 72°C for 15 sec

Repeat Step2-3 for 16-20 cycles

Keep at 4°C

1.4 Purify PCR products magnetic DNA purification beads as described in the manufacturers protocol:

- Gently shake AmpliClean beads to resuspend the magnetic particles to obtain a homogeneous suspension.
- Add 1.8 x volume of AmpliClean to the PCR product (e.g., 90 µl beads to 50 µl PCR)
- Mix thoroughly by gentle vortexing and incubate for 3 - 5 minutes at room temperature
- Place the reaction plate onto the Magnetic Ring Plate for 2 minutes.
- While sitting on the magnet, remove the cleared solution from the reaction plate and discard by pipetting from the center of the bottom of the wells. Make sure the removed solution is fully cleared and do not remove any magnetic particles.
- While leaving the plate on the magnet, immediately dispense 150 µL of 70% ethanol to each well of the reaction plate and incubate for 30 seconds at room temperature.
- Remove the ethanol and discard. Repeat this step and make sure to completely remove all ethanol with the last aspiration.
- Optional: Dry for max. 5 minutes at RT. Take plate from the magnet.
- While the plate is off the magnet, add 25 µL of elution buffer to each well of the reaction plate and homogenize the beads in the elution buffer by pipetting up and down.
- Place the reaction plate onto the magnet plate for 1 minute to separate beads from the solution and transfer the eluant, containing the purified PCR products to a new plate.
- Measure concentration of purified PCR product using Qbit HS assay.
- Run 10 µl of purified PCR product on gel to check product size (expected: 134 bp).

1.5 Set up golden gate reaction:

Compound	µl
Amplified Cas12a oligos	200 ng
pRDA_052	2 µg
2 x T7 Ligase buffer	34 µl
Esp3I	5,4 µl
T7 Ligase	2.7 µl
DTT	0.7 µl
BSA (10 mg/ml)	0.7 µl
H2O	to 68 µl

Note: Volumes and concentrations can be scaled up or down if necessary

- Incubate reaction in thermocycler:

Step1: 37°C ... 5 minutes
Step2: 20°C.... 5 minutes
Step3: 37°C.... 5 minutes
Repeat Step2 and 3 for 40 cycles
Keep at 4°C

- Purify golden gate reaction with 1 x volume AmpliClean beads as described above.
- Elute in 8 µl H₂O or EB buffer.

1.6 Transformation of plasmid library

Transformation is carried out in a 0.1 cm gap cuvette using 25 µL of Endura Electrocompetent Cells. Optimal settings for electroporation are listed in the table below. Typical time constants are 3.5 to 4.5 msec.

Electroporation settings:

- 1.0 mm cuvette
- 25 µF
- 200 Ohms
- 1500 Volts

Suggested Electroporation Systems:

Bio-Rad Micro Pulser #165-2100; Bio-Rad E. coli Pulser #165-2102; Bio-Rad Gene Pulser II #165-2105; BTX ECM630 Electroporation System; Eppendorf Electroporator 2510. Optional transformation control reactions include electroporation with 1 µL (10 pg) of supercoiled pUC19 DNA.

- Have Recovery Medium and 17 mm x 100 mm sterile culture tubes readily available at room temperature (one tube for each transformation reaction).
- Place electroporation cuvettes (0.1 cm gap) and microcentrifuge tubes on ice (one cuvette and one microfuge tube for each transformation reaction).
- Remove cells from the -80 °C freezer and place on wet ice until they thaw completely (10-20 minutes).
- When cells are thawed, mix them by tapping gently. Aliquot 2 x 25 µL of cells to the chilled microcentrifuge tubes on ice.
- Add 2 x 2.5 µl of purified golden gate reaction to each tube with 25 µl cells and mix by gently tapping the tube several times. Avoid air bubbles when pipetting.
- Carefully pipet 2 x 25 µL of the cell/DNA mixture into chilled electroporation cuvettes without introducing bubbles. Quickly flick the cuvette downward with your wrist to deposit the cells across the bottom of the well. Electroporate according to the conditions recommended above.
- Within 10 seconds of the pulse, add 975 µL of Recovery Medium to the cuvette and pipet up and down three times to resuspend the cells. Transfer the cells and Recovery Medium to a culture tube.
- Place the tube in a shaking incubator at 250 rpm for 1 hour at 37 °C and combine 2 x 1 ml afterwards.
- Spread 1 µl of undiluted and 1 µl of 1:10 diluted cells on agar plates with carbenicillin.
- Inoculate 250 ml ONCs with remaining cells (~2 ml).
- Incubate the plates overnight at 37°C

- Incubate 250 ml ONCs at 30°C over-night (preferably only 12-14 hours)
- After overnight incubation, count colonies on 0.1 µl plates and calculate total number of clones in ONC by 20.000.
- Usually, at least 50-fold coverage of the CRISPR library is recommended to achieve good representation of all gRNA combinations. Therefore, if the library contains 160.000 oligos, the 0.1 µl plate should contain at least 400 colonies.
- If coverage is substantially lower than 50x, repeat transformation with more competent cells.

1.7 Purify plasmid library and check quality.

- Harvest bacteria from 250 ml ONC and purify plasmid using Maxiprep kit (Qiagen).
- Measure plasmid concentration using nanodrop device.
- Send plasmid library for Sanger sequencing with hU6-f primer (GAGGGCCTATTTCCCATGATT) to Microsynth or another sequencing provider. Expect a sequence with double peaks at both gRNA sites that flank the DR sequence (Figure 1).

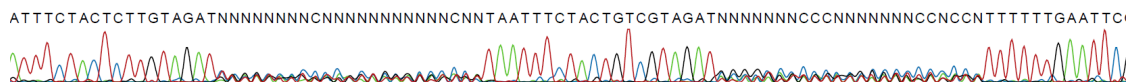


Figure 1: Sanger sequencing results for a typical combinatorial CRISPR/Cas12a library

- Pick 8-12 colonies from 0.1 µl plate and incubate 5 ml ONCs at 37°C.
- Next day, purify plasmids with a miniprep kit and sequence with hU6-f primer. >90% of plasmids should contain two gRNAs from the library.

Step 2: Transduction and cell culture

2.1 Generate enCas12a expressing clone (skip if Cas12a expressing clone is already available)

- Generate lentiviral particles using pRDA_174 vector (Addgene #136476) instead of library as described below (virus preparation).
- Transduce cell line of interest with lentiviral particles and select with appropriate concentration of blasticidin (to be tested for each cell line).
- Sort single cell clones from selected cell pool (by FACS or serial dilution) and expand clones.
- Test enCas12a activity of expanded clones using e.g. combinatorial gRNAs against two surface markers CD46 and CD81.
- You should reach at least 75% double KO efficiency in a clone before moving on to perform pooled genetic screens.

2.2 Virus preparation

Day 1:

- Seed (in the afternoon) 9×10^6 HEK 293T cells per plate in 2 x 15 cm plates (1x15 cm plate will yield 50 ml of supernatant with viral particles).

Day 2:

- Prepare transfection mix:

Component (per 15 cm plate)	µg	µl
Serum free medium		888
pMD2.G	5.83	
psPax2	8.74	
library	11.7	
PEI (stock: 1 mg/ml)		145
DMEM (w/o Pen/Strep)		2300

- Start the transfection in the morning.
- Mix the plasmids in the serum free medium.
- Add the PEI and vortex for 10 s, spin down if too much liquid stays on the side of the tube.
- Incubate for 15 min at RT.
- During the incubation, add 17 ml of fresh medium (w/o Pen/Strep) to the cells.
- Gently add medium (without Pen/Strep!) to the transfection mix and pipet up and down slowly 2x.
- Add dropwise 3.2 ml of the transfection mix per plate.
- Gently swirl the plate to distribute the mix.
- In the afternoon (7-8 hours after transfection): replace medium with 25 ml fresh harvest medium (without Pen/Strep!).

Day 3:

- In the afternoon, collect the virus medium from the plates in a falcon tube and store in the fridge.
- Add 25 ml fresh medium to the plates.

Day 4:

- In the afternoon, collect the virus medium from the plates and mix with the medium from the previous day.
- Filter the medium with a 0.45 µm filter.
- Freeze the medium at -80°C (freeze a few stocks in 1.5 ml tubes for titration).

2.3 Virus titration

Day 1:

- Seed 15.000 HCT116 cells/well in a 9-well plate, use medium without Pen/Strep.

Day 2:

- Thaw a virus stock from the -80° at RT.
- Add increasing amounts of virus to the cells (see below).
- Add Protamine Sulfate to each well (stock 8 mg/ml, final dilution: 1.3 µg/ml).

	1	2	3	4	5	6	7	8	9	10	11	12
	Dilution factor	200	100	50	30	20	10	5	3	ctrl 1	ctrl 2	
A	PBS	PBS										PBS
B		1.0	2.0	4.0	6.7	10.0	20.0	40.0	66.7	0	0	
C		1.0	2.0	4.0	6.7	10.0	20.0	40.0	66.7	0	0	
D		1.0	2.0	4.0	6.7	10.0	20.0	40.0	66.7	0	0	
E												
F												
G												
H		PBS										

Day 3:

- Transfer cells to 24-well plate in the morning, add selection medium (1 µg/ml Puromycin) in the afternoon.

Day 5 onwards:

- Wait until the selection control is dead (ctrl 1).
- If the no-selection control (ctrl 2) is confluent, transfer to a larger plate.
- Once the selection control is dead, quantify the percentage of live cells via Cell Titer Glow.
- Recommended: also count the cells with a cell counter as a control.
- For easy-to-transduce cell lines, a 1:100 dilution of the virus usually results in an MOI of ~0.3. If titer is significantly below, more viral particles need to be generated and possibly concentrated before transduction.

2.4 Transduction and selection

Day 1:

- Seed 400 million HCT116-enCas12a-c8 cells per 10-Layer stack.

Day 2:

- Replace medium with fresh medium (no Pen/Strep).
- Add virus at a dilution that resulted in an MOI of around 0.3 in the titration assay.
- Add 1.3 µg/ml protamine sulfate.

Day 3:

- Passage cells and re-seed 400 million cells per 10-Layer stack, also seed 9 million cells on 15 cm dishes to be able to observe them under the microscope.

In the afternoon, add puromycin to a final concentration of 1 µg/ml.

2.5 Passaging and cell harvesting

Materials needed per 10-Layer stack:

- 1.5 L RPMI (10% FBS, 1% P/S)
- 500 ml PBS
- 200 ml Trypsin

- A sterile plastic bottle with a capacity of >200 ml
- 3-7x 50 ml falcon tubes
- 2x15 ml falcon tubes
- CASY tube
- 10 ml CASYton
- Autoclaved glass bottles (total capacity needed: ~2.5 L)

General notes:

- Passage the cells 2 times per week:
 - For a 3-day incubation period, seed 250 million cells
 - For a 4-day incubation period, seed 220 million cells
- Once per week, collect 2x220 million cells per pool and freeze at -80°C.
- To distribute liquids equally across all layers:
 - Close both lids of the stack
 - Turn the stack onto its' long side and wait for the liquid to level out on all layers.
 - Turn the stack onto its' short side, away from the lids and then put the stack the right way up again.

Passaging:

- Pour medium from the 10-Layer stack into an autoclaved glass bottle.
- Wash with 250 ml PBS.
- Add 200 ml Trypsin.
- Incubate for 5 minutes.
- Once the cells have detached from the plate, pour them into a sterile bottle.
- Add 250 ml PBS to the stack to wash out cell clumps, leave the PBS while you are preparing the cells to avoid drying out any remaining cell clumps.
- Pour 200 ml medium into a second sterile bottle.
- Transfer 10 ml of the cells in trypsin into a 50 ml falcon tube, pipette up and down multiple times to loosen up any cell clumps, then transfer the cells to the bottle with medium.
- Once all cells have been transferred, transfer 1 ml to a safe-lock tube.
- Add 25 µl of the cell suspension to 10 ml of CASYton in a CASY tube.
- Count the cells using programme 19 on the CASY.
- Multiply the cell count by 4 and calculate the needed volume for reseeding the cells.
- Transfer the volume of cells to a bottle of fresh RPMI.
- Pour out the PBS from the 10-Layer stack, then pour the cell suspension into the stack.
- Add a total of 1.5 L RPMI to the stack.

Cell harvesting:

- Transfer 2x220 million cells to 50 ml falcon tubes.
- Centrifuge the cells at 300 g for 5 min.
- Remove the supernatant.
- Resuspend cell pellets in 2x10 ml PBS and transfer to 15 ml falcon tubes.

- Centrifuge the cells at 300 g for 5 min.

Resuspend the cells in 5 ml PBS with 20% DMSO then freeze at -80°C.

Step 3: gDNA purification

Materials needed:

- Qiagen genomic tips: 100/G (for up to 4×10^7 cells per sample) or 500/G (for up to 2×10^8 cells per sample)
- Qiagen Genomic DNA Buffer Set: C1, G2, QBT, QF and QC (can also be prepared in the lab)
- RNase A (100 mg/ml) (Qiagen)
- Proteinase K (20 mg/ml)
- TE buffer [pH 8.0]
- Optional: PacI enzyme (20 μ l per sample)

3.1 Prepare cell nuclei for purification:

- Use freshly harvested cells or thaw frozen cells on ice.
- Follow manufacturers protocol:
- Thaw frozen cells on ice and add 5 ml PBS.
- Centrifuge cells at 1500 x g for 10 min at 4°C.
- Remove supernatant and resuspend cells in 10 ml cold PBS. Divide cell suspension to 2 x 5 ml in 2 x 50 ml tubes. Add 5 ml cold PBS to each tube for a total of 10 ml volume.
- Add 10 ml ice cold buffer C1 and 30 ml ice cold distilled water to each 50 ml tube. Incubate on ice for 10 min.
- Centrifuge cells at 1300 x g for 15 min at 4°C (preferably in a swing-out rotor).
- Discard the supernatant and resuspend nuclei with 2 ml ice cold buffer C1 and 6 ml ice cold distilled water.
- Centrifuge cells at 1300 x g for 15 min at 4°C.
- Remove supernatant (leave around 300-500 μ l volume behind). And freeze nuclei at -80°C.

3.2 Purify gDNA from nuclei:

Genomic tips can be re-used to purify gDNA from another 1×10^8 cells from the same sample (equilibrate with buffer QBT after elution before next purification).

- Thaw one of the two tubes of each sample (1×10^8 cells).
- Resuspend nuclei in left-over volume of C1/H₂O volume (300-500 μ l) by vortexing at max. speed.
- Add 10 ml buffer G2 rapidly and pipet up and down a few times immediately to avoid precipitation. Vortex 10-30 sec. at max. speed.
- Add 20 μ l RNase A (100 mg/ml) to samples and mix by vortexing.

- Add 200 µl Proteinase K (20 mg/ml) to sample, mix by short vortexing and incubate at 50°C for 60 minutes.

Note: Incubation time may be extended if precipitates are still visible after 60 minutes. Sonication might aid to dissolve precipitates but leads to significant DNA degradation.

- Equilibrate a QIAGEN Genomic-tip 500/G with 10 ml of Buffer QBT and allow the QIAGEN Genomic-tip to empty by gravity flow.
- Vortex the sample for 10 s at maximum speed and apply it to the equilibrated QIAGEN Genomic-tip. Allow it to enter the resin by gravity flow.

Note: While the sample is flowing through column, start preparing the second tubes with nuclei for all samples (as above).

- Wash the QIAGEN Genomic-tip 2 x 15 ml of Buffer QC.
- Elute the genomic DNA with 1 x 15 ml of Buffer QF.
- Equilibrate used tip with 10 ml of Buffer QBT and apply other half of sample (1×10^8 cells). Repeat steps as before. In parallel, continue with DNA precipitation of first half of sample:
- Add 10.5 ml (0.7 volumes) room-temperature (15–25°C) isopropanol to the eluted DNA and precipitate the DNA by mixing vigorously and centrifuging immediately at $>5000 \times g$ for at least 15 min at 4°C. Carefully remove the supernatant.
- Wash the centrifuged DNA pellet with 4 ml of cold 70% ethanol. Vortex briefly and centrifuge at $>5000 \times g$ for 10 min at 4°C. Carefully remove the supernatant without disturbing the pellet. Air-dry for 5–10 minutes and resuspend the DNA in buffer.
- Dissolve the DNA in 750 µl TE buffer [pH 8.0] overnight on a shaker or at 55°C for 1–2 h. Resuspend the DNA pellet by rinsing the walls to recover the DNA.
- After elution of second half of sample, combine resuspended DNA (2 x 750 µl).
- Measure concentration of DNA sample. A total of 1200 µg of gDNA should be yielded to achieve full coverage for sequencing.

Note: If columns get clogged, a syringe with adapter can be used to apply pressure and assist flow-through.

3.3 (Optional): PacI digestion (for libraries cloned into pRDA_052_PacI vector)

The pRDA_052_PacI vector contains two PacI restriction site that were introduced directly upstream of the U6 promoter and 86 bp upstream of the EF1a promoter of pRDA_052. Digestion of gDNA containing pRDA_052 integration will result in 530 bp fragments that contain the gRNA site of the vector. This improves the efficiency of the subsequent amplification of the gRNA sites by PCR while leaving the remaining gDNA in large enough fragments to allow double size selection (PacI has an 8 bp recognition site that is not frequently present in the human genome).

- Set up PacI digestion (1800 µl volume):
 - 1500 µl gDNA (1200 µg)
 - 180 µl 10xCutSmart buffer
 - 20 µl PacI
 - 100 µl H₂O
- Digest at 37°C for 48 hours.
- Inactivate PacI enzyme at 65°C for 20 minutes.

Digested gDNA can be used for PCR without further purification but total volume of PCR should

be >5-times larger than digested gDNA volume to avoid buffer interference.

Step 4: Amplify combinatorial gRNAs for Illumina sequencing.

4.1 Primer sequences

P5 primer (modified from reference (2)):

```
pRDA_052_P5          AATGATACGGCGACCACCGAGATCTACACTTGTGGAAGGACGAAACACC*G
pRDA_052_P5_i5-D01   AATGATACGGCGACCACCGAGATCTACACAGGCTATACCTTGTGGAAGGACGAAACACC*G
pRDA_052_P5_i5-D02   AATGATACGGCGACCACCGAGATCTACACGCTCTATCTTGTGGAAGGACGAAACACC*G
```

Note: pRDA_052_P5_i5-D01 and pRDA_052_P5_i5-D02 contain i5 indices that can be used for dual indexing. However, i5 indices can only be read on MiSeq runs, not in NovaSeq runs since these primers do not contain the Illumina read1 sequence

P7 primer (modified from reference (2)):

```
GW_P7_1  CAAGCAGAAGACGGCATACGAGATCGGTTCAGTGAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_2  CAAGCAGAAGACGGCATACGAGATGCTGGATTGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_3  CAAGCAGAAGACGGCATACGAGATTAAGTGAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_4  CAAGCAGAAGACGGCATACGAGATTAACAGTGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_5  CAAGCAGAAGACGGCATACGAGATATACTCAAGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_6  CAAGCAGAAGACGGCATACGAGATGCTGAGAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_7  CAAGCAGAAGACGGCATACGAGATATTGGAGGGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_8  CAAGCAGAAGACGGCATACGAGATTAGTCTAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_9  CAAGCAGAAGACGGCATACGAGATCGGTGACCGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_10 CAAGCAGAAGACGGCATACGAGATTACAGAGGGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_11 CAAGCAGAAGACGGCATACGAGATATTGTCAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_12 CAAGCAGAAGACGGCATACGAGATTATGCTTGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_13 CAAGCAGAAGACGGCATACGAGATCGAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_14 CAAGCAGAAGACGGCATACGAGATGCCAAGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_15 CAAGCAGAAGACGGCATACGAGATATCCATGCGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_16 CAAGCAGAAGACGGCATACGAGATCATGATGCGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_17 CAAGCAGAAGACGGCATACGAGATAGACTGCGGTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
GW_P7_18 CAAGCAGAAGACGGCATACGAGATGACTGCATGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGGATGAATACTGCCATTTGTC*T
```

*: Phosphorothioate Bonds Modifications

Note: These are examples for 18 P7 primer with unique i7 indices that can be used for single Illumina indexing. Each sample that is intended to be multiplexed for sequencing needs to contain a unique i7 index. If more than 18 samples need to be multiplexed for NovaSeq sequencing, additional primer with unique indices need to be designed.

4.2 Set up PCR reactions to amplify combinatorial gRNAs

Set up test PCR to check template and PCR conditions:

	μl
5X Q5 buffer/Phusion HF buffer	10
10 μM primer (P5/P7)	2.5
NTPs (10 mM)	1
Q5 Pol	0.5
H2O	x

5 µg gDNA template	x
Total	50

- Run PCR program:
Step1: 98°C for 1 min
Step2: 98°C for 10 sec
Step3: 61°C for 30 sec
Step4: 72°C for 30 sec

Repeat Step2-4 for 5 cycles

Step5: 98°C for 10 sec
Step6: 72°C for 1 min

Repeat Step 5-6 for 19-22 cycles

Step7: 72°C for 2 min
Keep at 4°C
- Run PCR (10 µl) on agarose gel (expected PCR product size: 325 bp).
- If strong but not overamplified PCR (additional larger sized bands indicating "PCR bubbles" product, scale up PCRs and run tested program. E.g., for 1200 µg gDNA Input, set up 12000 µl PCR reaction and divide into 96 x 120 µl PCRs (expect some volume loss during pipetting) on a PCR plate.
- After PCR completion, combine 10 µl of each 120 µl PCR in a 1.5 ml tube.
- To generate plasmid libraries for sequencing (input controls): use 1-10 ng plasmid library as template per replicate, run program as above for 5 (61°C annealing) +10-15 (72°C annealing) cycles. Purify PCR product with 1,3 x volume Ampliclean beads as described above (no double size-selection required).

4.3 Double-size selection of pooled PCR products

- Gently shake the Ampliclean to resuspend the magnetic particles to obtain a homogeneous suspension.
- Add 0.65 x volume of Ampliclean to PCR product (e.g., 65 µl when using 100 µl of the pooled PCRs).
- Mix thoroughly by pipetting up and down and incubate for 3 - 5 minutes at room temperature.
- Place the reaction plate onto the Magnetic Ring Plate for 2 minutes or until solution becomes clear).
- While sitting on the magnet, transfer the cleared solution into new tube (this solution contains the unbound PCR product while large gDNA fragments are bound to beads).
- Add another 0.65 x (original) volume (e.g., 65 µl when 100 µl PCR were used) to the 165 µl cleared solution.
- Mix thoroughly by pipetting up and down and incubate for 3 - 5 minutes at room temperature.
- While sitting on the magnet, remove the cleared solution from the reaction plate and discard by pipetting from the center of the bottom of the wells. Make sure the removed solution is fully cleared and do not remove any magnetic particles.

- While leaving the plate on the magnet, immediately dispense 500 μ L of 70% ethanol to each well of the reaction plate and incubate for 30 seconds at room temperature.
- Remove the ethanol and discard. Repeat this step and make sure to completely remove all ethanol with the last aspiration.
- Optional: Dry for max. 5 minutes at RT. Take plate from the magnet.
- While plate is off the magnet, add 25 μ L of elution buffer to each well of the reaction plate and homogenize the beads in the elution buffer by pipetting up and down.
- Place the reaction plate onto the magnet plate for 1 minute to separate beads from the solution and transfer the eluant, containing the purified PCR products to a new plate.
- Measure concentration of purified PCR product using Qbit HS assay.
- Run 10 μ L of purified PCR product on gel to check product size and purity.

4.4 Multiplex libraries for Illumina sequencing

- Measure concentrations of purified PCRs using a Qbit DNA quantification system or a similar device.
- Multiplex libraries at equal amounts according to the measured concentration.
- The final multiplexed sample should contain at least 100 ng DNA at a concentration of >5 ng/μl. Exact requirements should be discussed with the corresponding sequencing facility.

Step 5: Library sequencing

5.1 Library sequencing

- The final Illumina library contains the following elements:

AATGATACGGCGACCACCGAGATCTACACTTGTGAAAGGACGAAACACCGGTAATTTCTACTCT
TGTAGATNNNNNNNNNNNNNNNNNNNNNNNNNTAATTTCTACTGTCGTAGATNNNNNNNNNNNNNN
NNNNNNNNNNNTTTTTTGAATTCGCTAGCAAGCTTGGCGTAACTAGATCTTGAGACAAATGGCAG
TATTCATCCAGATCGGAAGAGCACACGTCTGAACTCCAGTCACNNNNNNNNATCTCGTATGCCG
TCTTCTGCTTG

P5 adapter
Sequencing primer
gRNA1
DR
gRNA2
P7 adapter
P7 index

- Libraries need to be sequenced with a custom sequencing primer:

CTTGTGGAAAGGACGAAACACCGGTAATTTCTACTCTTGTAGA*T

- Chose sequencing platform to yield at least as many reads per sample to match read number to the number of clones in the stably transduced cell pool. E.g., when a library with 1×10^5 oligos was transduced into 3×10^8 cells at an MOI of 0.3 to reach 1×10^8

transduced clones at a 1000-fold coverage, aim for a minimum of 1.1×10^8 reads (considering that the read mapping rate is usually only 90-95%) per sample.

- Use 10% PhiX spike-in to improve colour balance, otherwise read quality will drop at the DR sequence.
- Sequence library in single-end mode for 66 (read1) + 8 (i7 index) cycles to cover both gRNA sequences.
- i7 indices for demultiplexing (reverse complement):

GW_P7_1	TTGAACCG
GW_P7_2	AATCCAGC
GW_P7_3	CCGAGTTA
GW_P7_4	AACTGTTA
GW_P7_5	TTGAGTAT
GW_P7_6	TTCTCAGC
GW_P7_7	CCTCCAAT
GW_P7_8	TTAGACTA
GW_P7_9	GGTCACCG
GW_P7_10	CCTCTGTA
GW_P7_11	TTGACAAT
GW_P7_12	AAGACATA
GW_P7_13	CGAGTTTCG
GW_P7_14	AGCTTGGC
GW_P7_15	GCATGGAT
GW_P7_16	GCATCATG
GW_P7_17	CGCAGTCT
GW_P7_18	ATGCAGTC

5.2 Data processing

- Check read quality using [FastQC](#), also freely available (no local installation required) on the Galaxy platform (4).
- Generate a tabular file with three columns that contains all gRNA combinations in the used library:

Column1: unique oligo name

Column2: unique oligo sequence (gRNA1-DR-gRNA2)

Column3: gene ID (gene-gene pair name), identical for all oligos targeting these two genes

- If read quality is low at the DR sequence, as shown in Figure 2, this region can be trimmed from the reads before mapping to the oligos in the library. In this case, simply remove the DR sequence from the mapping table to get the gRNA1-gRNA2 sequence.

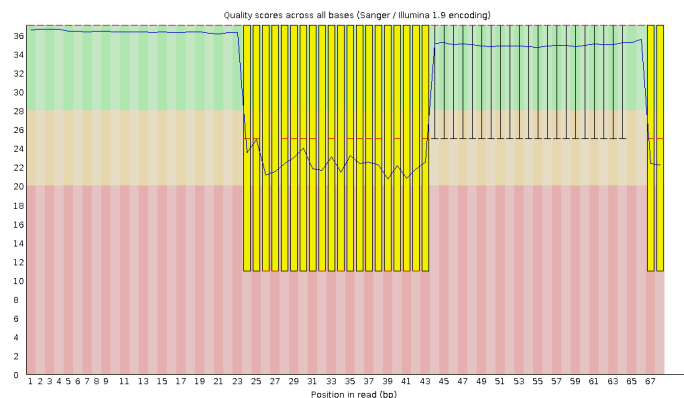


Figure 2: Per base sequence quality scores for a sequenced sample from a combinatorial CRISPR/enCas12a screen. Note that at position 24-43, where the DR sequence is located, the quality scores are significantly lower than at the two gRNA sequences (1-23 and 44-66).

- Use MAGeCK count tool (3) or custom script to map reads in fastq format to oligos. MAGeCK tools are also available on the Galaxy platform (4).
- The output tables contain raw and normalized read counts for each oligo.

Additional notes

Sections of this protocol were copied from manuals for reagents and kits as provided by the manufacturers. Some steps were modified to adapt to our workflow and should not be regarded as general recommendations for these reagents.

Data availability

Data from the SLC super family wide combinatorial CRISPR/enCas12a KO screen will be made available shortly at the [RESOLUTE database](#).

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